Chapter 2
Bayesian Inference

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Chapter 2. Bayesian Inference

Summarizing……

Bayesian Model

$$\begin{array}{l} \tilde{y} \mid \theta \sim p(\tilde{y} \mid \theta) \\ \theta \sim \pi(\theta) \end{array} \longrightarrow p(\tilde{y})$$

Data → Th. Bayes

Bayesian Model (updated)

$$\begin{array}{l} \tilde{y} \mid \theta \sim p(\tilde{y} \mid \theta) \\ \theta \mid y \sim \pi(\theta \mid y) \end{array} \longrightarrow p(\tilde{y} \mid y)$$

The updated model allows you to calculate point estimations, interval estimations, predictions…

Chapter 2. Bayesian Inference

After collecting the data and checking the model, Statistical Inference tries to guess the true parameter value, θ^*, that is, it tries to guess the model that generated the observed data,

$$p(\tilde{y} \mid \theta^*) \in M$$

Statistical Inference mainly consists of:

a) Point Estimation

b) Interval Estimation

c) Prediction

d) Hypothesis Test

Chapter 2. Bayesian Inference

2.1 Posterior distribution as an estimator

2.2 Point Estimation

2.3 Interval Estimation

2.4 Prediction

2.5 Hypothesis Test

2.1 Posterior Distribution as an Estimator

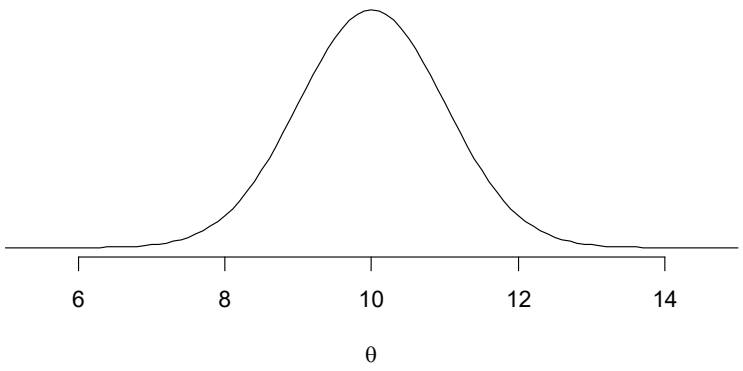
The posterior distribution, $\pi(\theta|y)$, has all the information about the parameter after observing the data,

$$\pi(\theta \mid y) = \frac{p(y \mid \theta)\pi(\theta)}{p(y)} \propto L_y(\theta)\pi(\theta)$$

The posterior distribution is a compromise between the prior distribution (the information before observing the data) and the likelihood function (the data information).

2.1 Posterior Distribution as an Estimator

The posterior distribution, $\pi(\theta|y)$, is the natural Bayesian estimator for θ



2.1 Posterior Distribution as an Estimator

Observation

We can do inference about $g(\theta)=\psi$ instead of θ:

a) Getting $\pi(\psi|y)$ analytically, or

b) Approximating $\pi(\psi|y)$ using simulations

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a) Getting $\pi(\psi|y)$ analytically:

$$\pi_\psi(\psi \mid y) = \pi_\theta(g^{-1}(\psi) \mid y)\left|\frac{\partial g^{-1}(\psi)}{\partial \psi}\right|$$

2.1 Posterior Distribution as an Estimator

b) Approximating $\pi(\psi|y)$ by simulation

To approximate $\pi(\psi|y)$ by simulation, first of all, we must set the number of simulations, M, (the larger the better) then from $j=1$ to M:

1. Simulate $\theta^{(j)}$ from $\pi(\theta \mid y)$
2. calculate $\psi^{(j)} = g(\theta^{(j)})$

These $\psi^{(1)},\ldots,\psi^{(M)}$ simulated values are simulations from the $\pi(\psi|y)$. Hence, using these simulations we can calculate everything we want/need (moments, probabilities, etc.). And by graphing it we can reproduce the shape of its probability distribution.

Chapter 2. Bayesian Inference

2.1 Posterior distribution as an estimator

2.2 Point Estimation

2.3 Interval Estimation

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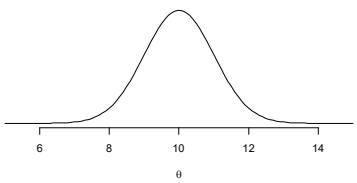
2.5 Hypothesis Test

2.2 Point Estimation

A Point estimator summarizes $\pi(\theta|y)$



It consists of choosing $\hat{\theta}$ in such a way

That it is the best approximation to θ^*

that we know how to do, using all the information we have

2.2 Point Estimation

Any measure of the location of $\pi(\theta|y)$ will serve as a point estimate:

- $\hat{\theta}_{ep} = E(\theta \mid y) = \int \theta \pi(\theta \mid y) \partial\theta$
- $\hat{\theta}_{mep}$ is such that $\int_{-\infty}^{\hat{\theta}_{mep}} \pi(\theta \mid y) \partial\theta = 0.5$
- $\hat{\theta}_{mop}$ is such that it maximizes $\pi(\theta \mid y)$

Observation: we can define a prior point estimator using the prior distribution instead of the posterior distribution

2.2 Point Estimation

Any measure of the location of $\pi(\theta|y)$ will serve as a point estimate:

- $\hat{\theta}_{ep} = E(\theta \mid y) = \int \theta \pi(\theta \mid y) \partial\theta$
- $\hat{\theta}_{mep}$ is such that $\int_{-\infty}^{\hat{\theta}_{mep}} \pi(\theta \mid y) \partial\theta = 0.5$
- $\hat{\theta}_{mop}$ is such that it maximizes $\pi(\theta \mid y)$

Choosing one or the other estimator is equivalent to choosing a specific loss function and choosing the estimator that minimizes it.

2.2 Point Estimation

Example

If we choose the quadratic loss function, then:

$$E((\theta-\hat{\theta})^2 \mid y) = \int (\theta-\hat{\theta})^2 \pi(\theta \mid y)\partial\theta = \int (\theta^2 - \hat{\theta}^2 - 2\theta\hat{\theta})\pi(\theta \mid y)\partial\theta =$$
$$= \int \theta^2 \pi(\theta \mid y)\partial\theta + \hat{\theta}^2 \int \pi(\theta \mid y)\partial\theta - 2\hat{\theta}\int \theta\pi(\theta \mid y)\partial\theta =$$
$$= E(\theta^2 \mid y) + \hat{\theta}^2 - 2\hat{\theta}E(\theta \mid y)$$

and deriving this expression with respect to $\hat{\theta}$ and equalizing it to zero we get that the value that minimizes the quadratic loss function is:

$$\hat{\theta} = E(\theta \mid y)$$

2.2 Point Estimation

Observation: We can write the posterior expected value as

$$E(\theta \mid y) = \lambda\hat{\theta}_{MV} + (1-\lambda)E(\theta), \quad \lambda \in [0,1]$$

that is, the posterior expected value is a weighed average between the Maximum Likelihood Estimator and the expected prior value.

The larger λ, the more weight the likelihood will have, and this will happen when the sample size is larger

The lower λ, the more weight the prior will have and this will happen when the prior is more informative, that is, it has less variance.

2.2 Point Estimation

Example

Given the conjugate Bayesian model, and observed data n , y

$$\tilde{y} \mid \theta \sim Binomial(n, \theta)$$
$$\theta \sim Beta(a, b)$$
$$\theta \mid y \sim Beta(a+y, b+n-y)$$

$$E(\theta \mid y) = \frac{a+y}{a+b+n} = \frac{y}{a+b+n} + \frac{a}{a+b+n} =$$

Simple size

Sample size equivalent to the prior information

$$= \left(\frac{n}{a+b+n}\right)\frac{y}{n} + \left(\frac{a+b}{a+b+n}\right)\frac{a}{a+b} =$$

$$= \lambda \hat{\theta}_{MV} + (1-\lambda)E(\theta), \quad \lambda = \left(\frac{n}{a+b+n}\right)$$

2.2 Point Estimation

Observation

We can get the point estimates:

a) analytically

b) by simulation

2.2 Point Estimation

Example: The posterior expected value

a) Analytically

$$\hat{\theta}_{ep} = E(\theta \mid y) = \int \theta \pi(\theta \mid y) \partial \theta$$

b) By simulation

Be $\theta^{(1)}, \ldots, \theta^{(M)}$ simulations of $\pi(\theta \mid y)$ then

$$\hat{\theta}_{ep} \approx \frac{\sum_{j=1}^{M} \theta^{(j)}}{M}$$

The larger M (the number of simulations), the better the approximation

Chapter 2. Bayesian Inference

2.1 Posterior distribution as an estimator

2.2 Point Estimation

2.3 Interval Estimation

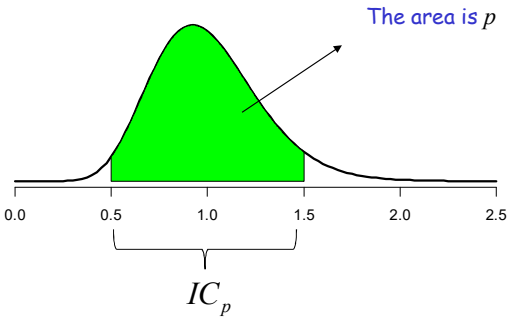
2.4 Prediction

2.5 Hypothesis Test

2.3 Interval Estimation

A posterior credibility (or probability) interval of p for θ , IC_p, is any region in Ω in such a way that

$$p(\theta \in IC_p \mid y) = \int_{IC_p} \pi(\theta \mid y) \partial\theta = p$$



Observation: we can define in an similar way a prior IC_p using the prior distribution instead of the posterior distribution.

2.3 Interval Estimation

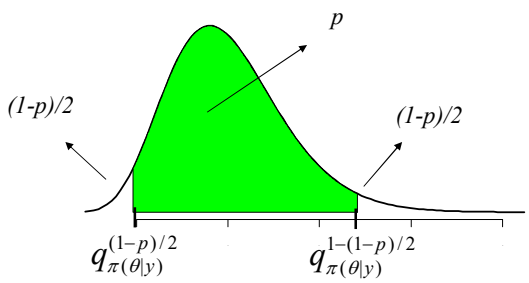
There are two main types of credible intervals:

a) Intervals based on percentiles

b) Highest posterior density intervals

2.3 Interval Estimation

a) Intervals based on percentiles



2.3 Interval Estimation

Observation

The Credible Intervals based on percentiles can be calculatee in two ways:

a) Analytically

b) By simulation

2.3 Interval Estimation

Example: A 95% Credible Interval for θ

a) Analytically

$$\int_{-\infty}^{q^{0.025}_{\pi(\theta|y)}} \pi(\theta \mid y)\partial\theta = 0.025$$

$$\int_{q^{0.975}_{\pi(\theta|y)}}^{\infty} \pi(\theta \mid y)\partial\theta = 0.025$$

The larger M (the number of simulations) the better the approximation

b) By simulation

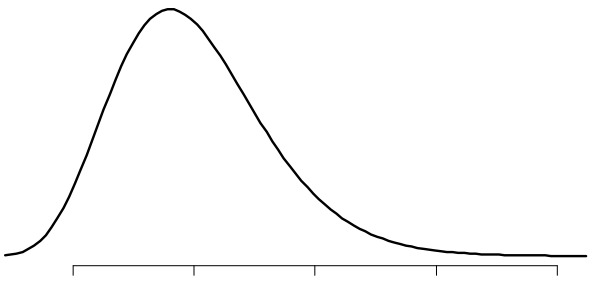
Be $\theta^{(1)},\ldots,\theta^{(M)}$ simulations of $\pi(\theta \mid y)$ then

$$\left(q^{0.025}_{\theta^{(1)},\ldots,\theta^{(M)}},\ q^{0.975}_{\theta^{(1)},\ldots\theta^{(M)}}\right)$$

percentile 2.5% percentile 97.5%

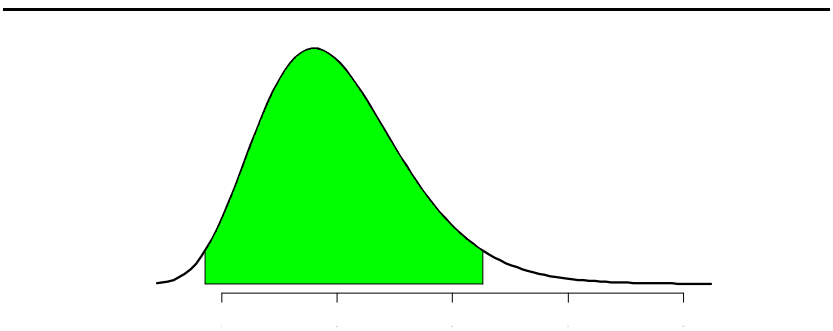
2.3 Interval Estimation

b) Highest density Interval



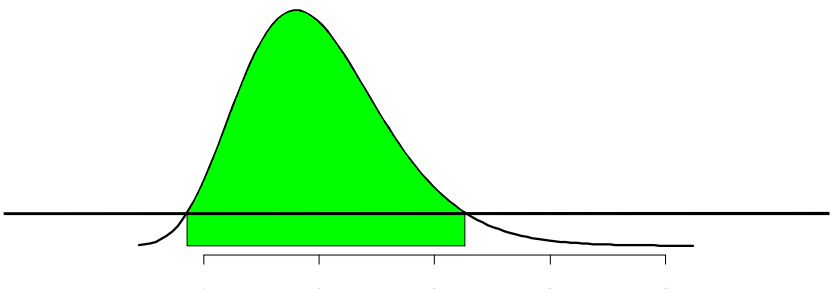
2.3 Interval Estimation

b) Highest density Interval



2.3 Interval Estimation

b) Highest density Interval



They are the narrowest possible intervals

2.3 Interval Estimation

Observation

We can do inference about $g(\theta)=\psi$ instead of θ:

a) Getting $\pi(\psi|y)$ analytically, or

b) Approximating $\pi(\psi|y)$ using simulations

---

a) Getting $\pi(\psi|y)$ analytically:

$$\pi_\psi(\psi \mid y) = \pi_\theta(g^{-1}(\psi) \mid y)\left|\frac{\partial g^{-1}(\psi)}{\partial \psi}\right|$$

2.3 Interval Estimation

b) Approximating $\pi(\psi|y)$ by simulation

To approximate $\pi(\psi|y)$ by simulation, first of all, we must set the number of simulations, M, (the larger the better) then from $j=1$ to M:

1. Simulate $\theta^{(j)}$ from $\pi(\theta \mid y)$
2. calculate $\psi^{(j)} = g(\theta^{(j)})$

These $\psi^{(1)},\ldots,\psi^{(M)}$ simulated values are simulations from the $\pi(\psi|y)$. Hence, using these simulations we can calculate everything we want/need (moments, probabilities, etc.). And by graphing it we can reproduce the shape of its probability distribution.

2.3 Interval Estimation

Now, using these simulated values: $\psi^{(1)},\ldots,\psi^{(M)}$

An estimator for ψ could be:

$$\widehat{\psi} = E(\psi \mid y) \approx \frac{\sum_{j=1}^{M} \psi^{(j)}}{M}$$

And, a 95% credible interval for ψ could be:

$$\left(q^{0.025}_{\psi^{(1)},\ldots,\psi^{(M)}},\ q^{0.975}_{\psi^{(1)},\ldots,\psi^{(M)}}\right)$$

percentile 2.5% percentile 97.5%

Chapter 2. Bayesian Inference

2.1 Posterior distribution as an estimator

2.2 Point Estimation

2.3 Interval Estimation

2.4 Prediction

2.5 Hypothesis Test

2.4 Prediction

In a similar way the posterior distribution,

$\pi(\theta \mid y)$ represents all that we know about the parameter

The posterior predictive distribution,

$p(\tilde{y} \mid y)$ represents all that we know about future values for $\tilde{y}$

2.4 Prediction

The posterior predictive distribution can be calculated in two ways:

a) Analytically

b) By simulation

---

a) Analytically:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta)\pi(\theta \mid y)d\theta$$

2.4 Prediction

To approximate the posterior predictive distribution using simulations, first of all, you must set the number of simulations, *M* (the larger the better), then for *j*=1 to *M*:

1. Simulate

$$\theta^{(j)} \text{ de } \pi(\theta \mid y), \text{ and}$$

2. Simulate

$$\tilde{y}^{(j)} \text{ de } p(\tilde{y} \mid \theta^{(j)})$$

These $\tilde{y}^{(1)}, \ldots, \tilde{y}^{(M)}$ simulated values are simulations from the posterior predictive distribution. Hence, using these simulations we can calculate everything we want/need (moments, probabilities, etc.). And, by graphing it we can reproduce the shape of its probability distribution.

2.4 Prediction

We can also calculate point estimates and credibility intervals for the predictions. We can calculate them analytically or by simulation.

Example:

a) Analytically:

- $E(\tilde{y} \mid y) = \int \tilde{y} p(\tilde{y} \mid y) \partial \tilde{y} = \int \tilde{y} \int p(\tilde{y} \mid \theta) \pi(\theta \mid y) d\theta \partial \tilde{y}$

- $CI_{95\%} = (q^{0.025}_{p(\tilde{y}|y)}, q^{0.975}_{p(\tilde{y}|y)})$

  Where $q^{0.025}_{p(\tilde{y}|y)}$ $q^{0.975}_{p(\tilde{y}|y)}$ are such that,

$$\int_{-\infty}^{q^{0.025}_{p(\tilde{y}|y)}} p(\tilde{y} \mid y) \partial \theta = 0.025, \quad \int_{q^{0.975}_{p(\tilde{y}|y)}}^{\infty} \pi(\theta \mid y) \partial \theta = 0.025$$

2.4 Prediction

Example: point estimate and credibility interval for a prediction

b) By simulation

Be $\tilde{y}^{(1)},\ldots,\tilde{y}^{(M)}$ simulations from the posterior predictive distribution

- $E(\tilde{y}\mid y)\approx\dfrac{\sum_{j=1}^{M}\tilde{y}^{(j)}}{M}$

- $IC_{95\%}=\left(q^{0.025}_{\tilde{y}^{(1)},\ldots,\tilde{y}^{(M)}},\ q^{0.975}_{\tilde{y}^{(1)},\ldots,\tilde{y}^{(M)}}\right)$

percentile 2.5% percentile 97.5%

Chapter 2. Bayesian Inference

2.1 Posterior distribution as an estimator
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2.5 Hypothesis Test

2.5 Hypothesis Test

The main idea

We start from a Bayesian Model,

$$M = \{p(\tilde{y} \mid \theta), \theta \in \Omega\}, \quad \pi(\theta)$$

We split the parameter space into two or more supspaces. Then, we want to answer the question: "To which supspace does the real value of the parameter (θ^*) belong?".

2.5 Hypothesis Test

Example 1

Given a Bayesian Model: $M = \{p(\tilde{y} \mid \theta), \theta \in \Omega\}, \quad \pi(\theta)$

We split the parameter space in two: $\Omega = \Omega_1 \cup \Omega_2$

We want to decide to which subspace the parameter belongs

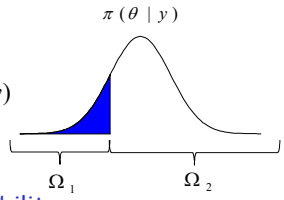
$H_1 : \theta \in \Omega_1$

$H_2 : \theta \in \Omega_2$

After observing the data, we update the Bayesian Model and compute the posterior distribution for each hypothesis as:

$$p(H_1 \mid y) = p(\theta \in \Omega_1 \mid y) = \int_{\Omega_1} \pi(\theta \mid y) d\theta$$

$$p(H_2 \mid y) = p(\theta \in \Omega_2 \mid y) = \int_{\Omega_2} \pi(\theta \mid y) d\theta = 1 - p(H_1 \mid y)$$



We choose the hypothesis with the highest probability

2.5 Hypothesis Test

Hypothesis test and model selection

Splitting the parameter space is equivalent to splitting the model M into submodels

$$M = \{p(\tilde{y} \mid \theta), \theta \in \Omega\}; \quad \Omega = \Omega_1 \bigcup \Omega_2$$

$$\left.\begin{array}{l} M_1 = \{p(\tilde{y} \mid \theta), \theta \in \Omega_1\} \\ M_2 = \{p(\tilde{y} \mid \theta), \theta \in \Omega_2\} \end{array}\right\} \Rightarrow M = M_1 \bigcup M_2$$

Performing a hypothesis test is equivalent to making a model selection.

$$H_1 : \theta \in \Omega_1 \quad \Rightarrow \quad M_1 = \{p(\tilde{y} \mid \theta), \theta \in \Omega_1\}$$
$$H_2 : \theta \in \Omega_2 \quad \Rightarrow \quad M_2 = \{p(\tilde{y} \mid \theta), \theta \in \Omega_2\}$$

$$p(M_1 \mid y) = p(H_1 \mid y) = p(\theta \in \Omega_1 \mid y) = \int_{\Omega_1} \pi(\theta \mid y) d\theta$$
$$p(M_2 \mid y) = p(H_2 \mid y) = p(\theta \in \Omega_2 \mid y) = \int_{\Omega_2} \pi(\theta \mid y) d\theta$$

2.5 Hypothesis Test

Case: Two possible values (Simple vs. Simple)

The parameter space is discrete and has only two possible values.

$$M = \{p(\tilde{y} \mid \theta), \theta \in \{\theta_1, \theta_2\}\} \qquad \pi(\theta) = \begin{cases} p & si \quad \theta = \theta_1 \\ 1-p & si \quad \theta = \theta_2 \end{cases}$$

$$H_1 : \theta = \theta_1 \quad \Rightarrow \quad p(H_1) = p$$
$$H_2 : \theta = \theta_2 \quad \Rightarrow \quad p(H_2) = 1 - p$$

We calculate the posterior distribution of each hypothesis as:

$$p(H_1 \mid y) = \frac{p(H_1) p(y \mid H_1)}{p(y)} = \frac{p(H_1) p(y \mid H_1)}{p(H_1) p(y \mid H_1) + p(H_2) p(y \mid H_2)}$$

$$p(H_2 \mid y) = 1 - p(H_1 \mid y)$$

We choose the hypothesis with the highest probability

Observation: $$\pi(\theta \mid y) = \begin{cases} p(H_1 \mid y) & si \quad \theta = \theta_1 \\ p(H_2 \mid y) & si \quad \theta = \theta_2 \end{cases}$$

2.5 Hypothesis Test

Example 2: Two possible values

We have a manipulated coin, with a probability of 0.7 of getting head or tail (this is what we want to know). We will toss the coin 10 times and count the number of observed heads.

$$M = \{p(\tilde{y} \mid \theta) = Binomial(n = 10, \theta), \theta \in \{0.3, 0.7\}\}$$
$$M = \{Binomial(n = 10, \theta = 0.3), Binomial(n = 10, \theta = 0.7)\}$$

$$H_1 : \theta = 0.3 \qquad \pi(\theta) = \begin{cases} 0.5 & si \quad \theta = 0.3 \\ 0.5 & si \quad \theta = 0.7 \end{cases}$$
$$H_2 : \theta = 0.7$$

We toss the coin $n=10$ times and observe $y=7$ heads.

Now we have to calculate the posterior probability of each hypothesis.

2.5 Hypothesis Test

Example 2 (continued): Two possible values

$$M = \{p(\tilde{y} \mid \theta) = Binomial(n = 10, \theta), \theta \in \{0.3, 0.7\}\} \qquad \pi(\theta) = \begin{cases} 0.5 & si \quad \theta = 0.3 \\ 0.5 & si \quad \theta = 0.7 \end{cases}$$

$$H_1 : \theta = 0.3$$
$$H_2 : \theta = 0.7$$

We toss the coin $n=10$ times and observe $y=7$ heads.

$$p(H_1 \mid y) = \frac{p(H_1)p(y \mid H_1)}{p(y)} = \frac{p(H_1)p(y \mid H_1)}{p(H_1)p(y \mid H_1) + p(H_2)p(y \mid H_2)} =$$

$$= \frac{0.5 \times \binom{10}{7} 0.3^7 (1 - 0.3)^{10-7}}{0.5 \times \binom{10}{7} 0.3^7 (1 - 0.3)^{10-7} + 0.5 \times \binom{10}{7} 0.7^7 (1 - 0.7)^{10-7}} = 0.033$$

$$p(H_2 \mid y) = 1 - p(H_1 \mid y) = 0.967$$

We choose H_2

2.5 Hypothesis Test

Bayes Factor

Given two hypothesis, H_1 and H_2, we can express the posterior odds as follows:

$$\frac{p(H_1 \mid y)}{p(H_2 \mid y)} = \frac{\dfrac{p(H_1)p(y \mid H_1)}{p(y)}}{\dfrac{p(H_2)p(y \mid H_2)}{p(y)}} = \frac{p(H_1)}{p(H_2)} \frac{p(y \mid H_1)}{p(y \mid H_2)}$$

$= BF_{12}$

Observation: $p(y) = \sum_{i=1}^{2} p(H_i)p(y \mid H_i) = p(H_1)p(y \mid H_1) + p(H_2)p(y \mid H_2)$

Posteriori Odds = Priori Odds x Bayes Factor

Hence, we can also calculate the Bayes Factor as:

$$FB_{12} = \frac{p(H_2)}{p(H_1)} \frac{p(H_1 \mid y)}{p(H_2 \mid y)}$$

2.5 Hypothesis Test

Bayes Factor

Given the two hypothesis, H_1 and H_2, if:

$$\frac{p(H_1 \mid y)}{p(H_2 \mid y)} > 1 \Leftrightarrow p(H_1 \mid y) > p(H_2 \mid y)$$ We choose H_1

The Bayes factor measures the support of the Data for a model,

$$\frac{p(H_1 \mid y)}{p(H_2 \mid y)} = \frac{p(H_1)}{p(H_2)} \frac{p(y \mid H_1)}{p(y \mid H_2)}$$

$= FB_{12}$

Therefore, if

$$FB_{12} = \begin{cases} > 1 & \text{Data evidence in favor of hypothesis 1} \\ < 1 & \text{Data evidence in favor of hypothesis 2} \end{cases}$$

2.5 Hypothesis Test

Example 2 (continued): Two possible values

$$M = \{p(\tilde{y} \mid \theta) = Binomial(n = 10, \theta), \theta \in \{0.3, 0.7\}\} \qquad \pi(\theta) = \begin{cases} 0.5 & si \quad \theta = 0.3 \\ 0.5 & si \quad \theta = 0.7 \end{cases}$$

$$H_1 : \theta = 0.3$$
$$H_2 : \theta = 0.7$$

We toss a coin $n=10$ times and observe $y=7$ heads.

$$p(H_1) = p(H_2) = 0.5 \quad \Rightarrow \quad odds\ a\ priori = \frac{p(H_1)}{p(H_2)} = 1$$

$$\frac{p(H_1 \mid y)}{p(H_2 \mid y)} = 1 \times FB_{12} = \frac{p(y \mid H_1)}{p(y \mid H_2)} = \frac{\binom{10}{7} 0.3^7 (1-0.3)^{10-7}}{\binom{10}{7} 0.7^7 (1-0.7)^{10-7}} = \frac{0.009}{0.267} = 0.034$$

$$FB_{12} = 0.034$$
$$FB_{21} = \frac{1}{FB_{12}} = 29.4$$

=> Data evidence in favor of hypothesis 2

2.5 Hypothesis Test

Factor de Bayes

Kass and Raftery (1995)* propose a scale to help the interpretation of the BF:

$\log_{10}$ FB	FB	Strength of evidence
0 to 1/2	1 to 3.2	Not worth more than a bare mention
1/2 to 1	3.2 to 10	Substantial
1 to 2	10 to 100	Strong
> 2	> 100	Decisive

*Robert E. Kass & Adrian E. Raftery (1995). "Bayes Factors" (PDF). *Journal of the American Statistical Association*. 90 (430): 791. doi:10.2307/2291091. JSTOR 2291091. http://www.andrew.cmu.edu/user/kk3n/simplicity/KassRaftery1995.pdf

2.5 Hypothesis Test

Example 3: Three hypotheses (Composite vs. Composite)

The time needed for a specific radioactive particle to disintegrate follows an *Exponencial* Model. Physicists agree to use a *Gamma*(*a*=10, *b*=10) as a prior distribution.

The Bayesian model is:

$$\tilde{y} \mid \lambda \sim \exp(\lambda)$$
$$\lambda \sim Gamma(10, 10)$$

$$p(y|\lambda) = \lambda e^{-\lambda y}$$
$$\pi(\lambda) = \frac{b^a \lambda^{(a-1)} e^{-b\lambda}}{\Gamma(a)}$$

To avoid errors related to the parametrizations, it is a good idea to show the expressions for the distributions.

We want to chose among:

$$H_1: \ \lambda \in (0, 0.5)$$
$$H_2: \ \lambda \in [0.5, 1.5)$$
$$H_3: \ \lambda \in [1.5, \infty)$$

2.5 Hypothesis Test

Example 3: Three hypotheses (continued)

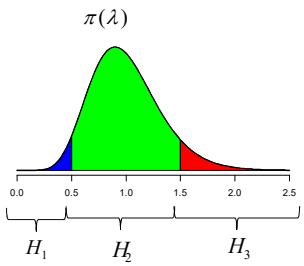
Given the prior distribution and the hypothesis to test:

$$\lambda \sim Gamma(10, 10)$$

$$H_1: \ \lambda \in (0, 0.5)$$
$$H_2: \ \lambda \in [0.5, 1.5)$$
$$H_3: \ \lambda \in [1.5, \infty)$$

We can calculate the prior probability for each hypothesis as:

$$p(H_1) = \int_0^{0.5} \pi(\lambda) d\lambda = 0.032$$
$$p(H_2) = \int_{0.5}^{1.5} \pi(\lambda) d\lambda = 0.898$$
$$p(H_3) = \int_{1.5}^{\infty} \pi(\lambda) d\lambda = 0.070$$



2.5 Hypothesis Test

Example 3: Three hypotheses (continued)

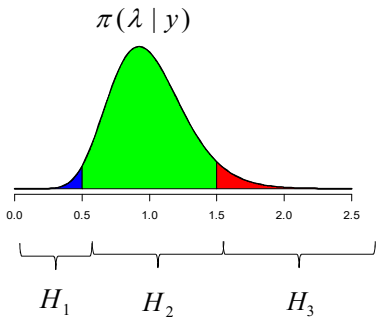
The observed data is: 0.9, 1.1 and 1.

The posterior distribution is: $\lambda \mid y \sim Gamma(13, 13)$

We calculate the posterior distribution for each hypothesis as follows:

$$H_1: \ \lambda \in (0, 0.5)$$
$$H_2: \ \lambda \in [0.5, 1.5)$$
$$H_3: \ \lambda \in [1.5, \infty)$$

$$p(H_1 \mid y) = \int_0^{0.5} \pi(\lambda \mid y) d\lambda = 0.016$$
$$p(H_2 \mid y) = \int_{0.5}^{1.5} \pi(\lambda \mid y) d\lambda = 0.935$$
$$p(H_3 \mid y) = \int_{1.5}^{\infty} \pi(\lambda \mid y) d\lambda = 0.049$$



We choose H_2

2.5 Hypothesis Test

Bayes Factor

We can calculate the Bayes Factor of hypothesis *i* against hypothesis *j*

$$FB_{ij} = \frac{p(H_j)}{p(H_i)} \frac{p(H_i \mid y)}{p(H_j \mid y)}$$

We can also calculate the Bayes Factor of the hypothesis *i* against all the other hypotheses

$$FB_{i.} = \frac{1 - p(H_i)}{p(H_i)} \frac{p(H_i \mid y)}{1 - p(H_i \mid y)}$$

2.5 Hypothesis Test

Example 3: Three hypotheses (continued)

The Bayes Factor for each pair of hypotheses, FB_{ij}, is:

i \ j	1	2	3
1	-	0.48	0.71
2	2.08	-	1.48
3	1.4	0.67	-

$$FB_{23} = \frac{p(H_3)}{p(H_2)}\frac{p(H_2 \mid y)}{p(H_3 \mid y)} = \frac{0.07}{0.898}\frac{0.935}{0.049} = 1.48$$

Data gives more evidence in favor of hypothesis 2 than hypothesis 3

The Bayes Factor for each hypothesis against all the other hypotheses is:

$$FB_{1.} = 0.492$$
$$FB_{2.} = 1.634$$
$$FB_{3.} = 0.685$$

Data gives more evidence in favor of hypothesis 2

2.5 Hypothesis Test

Example 4: Simple vs Composite

We want to find out if a coin is manipulaed or not. For this reason we're going to toss the coin n times and count the number of heads observed.

Given the Bayesian model:

$$M = \{p(y \mid \theta) = Binomial(n, \theta), \theta \in (0,1)\}; \quad \pi(\theta) = Beta(1,1)$$

We want to carry out this hypothesis test:

$$H_1 : \theta = 0.5$$
$$H_2 : \theta \neq 0.5$$

2.5 Hypothesis Test

Hypothesis Test and Model Selection

Splitting the parameter space is equivalent to splitting the model M into submodels

$$M = \{p(\tilde{y} \mid \theta), \theta \in \Omega\}; \quad \Omega = \Omega_1 \bigcup \Omega_2$$

$$\left.\begin{array}{l} M_1 = \{p(\tilde{y} \mid \theta), \theta \in \Omega_1\} \\ M_2 = \{p(\tilde{y} \mid \theta), \theta \in \Omega_2\} \end{array}\right\} \Rightarrow M = M_1 \bigcup M_2$$

Therefore, performing a hypothesis test is equivalent to selecting a model

$$H_1 : \theta \in \Omega_1 \quad \Rightarrow \quad M_1 = \{p(\tilde{y} \mid \theta), \theta \in \Omega_1\}$$
$$H_2 : \theta \in \Omega_2 \quad \Rightarrow \quad M_2 = \{p(\tilde{y} \mid \theta), \theta \in \Omega_2\}$$

$$p(M_1 \mid y) = p(H_1 \mid y) = p(\theta \in \Omega_1 \mid y) = \int_{\Omega_1} \pi(\theta \mid y) d\theta$$
$$p(M_2 \mid y) = p(H_2 \mid y) = p(\theta \in \Omega_2 \mid y) = \int_{\Omega_2} \pi(\theta \mid y) d\theta$$

2.5 Hypothesis Test

Hypothesis Test : the most general case

The hypothesis test seen before is a particular case of this more general approach:

$$H_1 : M_1 = \{p_1(\tilde{y} \mid \theta_1), \theta_1 \in \Omega_1\}; \quad \pi_1(\theta_1); \quad p(H_1) = p(M_1)$$
$$H_2 : M_2 = \{p_2(\tilde{y} \mid \theta_2), \theta_2 \in \Omega_2\}; \pi_2(\theta_2); \quad p(H_2) = p(M_2)$$
$$\vdots$$
$$H_k : M_k = \{p_k(\tilde{y} \mid \theta_k), \theta_k \in \Omega_k\}; \pi_k(\theta_k); \quad p(H_k) = p(M_k)$$

2.5 Hypothesis Test

Hypothesis Test : the most general case

For each model we have:

$$H_i : M_i = \{p_i(\tilde{y} \mid \theta_i), \theta_i \in \Omega_i\}; \pi_i(\theta_i); \quad p(H_i) = p(M_i)$$

where,

$$M_i = \{p_i(\tilde{y} \mid \theta_i), \theta_i \in \Omega_i\}$$

Each model is a list of probability distributions with its own parameters and parameter space.

$$\pi_i(\theta_i) = \begin{cases} \pi_i(\theta_i) & si\ \theta_i \in \Omega_i \\ 0 & otherwise \end{cases}$$

The prior distribution in each model is defined in its own parameter space.

$$p(H_i) = p(M_i)$$

The prior probability for each model (hypothesis) must be defined.

$$p(\tilde{y} \mid M_i) = \int_{\Omega_i} p_i(\tilde{y} \mid \theta_i)\pi_i(\theta_i)d\theta_i$$

Each model has its own prior predictive distribution

2.5 Hypothesis Test

Hypothesis Test : the most general case

Given k models:

$$H_1 : M_1 = \{p_1(\tilde{y} \mid \theta_1), \theta_1 \in \Omega_1\}; \quad \pi_1(\theta_1); \quad p(H_1) = p(M_1)$$
$$H_2 : M_2 = \{p_2(\tilde{y} \mid \theta_2), \theta_2 \in \Omega_2\}; \pi_2(\theta_2); \quad p(H_2) = p(M_2)$$
$$\vdots$$
$$H_k : M_k = \{p_k(\tilde{y} \mid \theta_k), \theta_k \in \Omega_k\}; \pi_k(\theta_k); \quad p(H_k) = p(M_k)$$

We compute the posterior probability for each Model (Hypothesis) as:

$$p(M_i \mid y) = \frac{p(M_i)p(y \mid M_i)}{\sum_{j=1}^{k} p(M_j)p(y \mid M_j)}$$

where:

$$p(y \mid M_i) = \int_{\Omega_i} p_i(y \mid \theta_i)\pi_i(\theta_i)d\theta_i$$

2.5 Hypothesis Test

Example 4: Simple vs Composite (continued)

We want to find out if a coin is manipulated or not. We're going to toss the coin n times and count the number of heads observed. And, we will perform the next Hypothesis test:

$$H_1 : \theta = 0.5$$
$$H_2 : \theta \neq 0.5$$

We can define the hypothesis test by defining the following models:

$$H_1 : M_1 = \{p(\tilde{y} \mid \theta) = Binomial(n, \theta = 0.5)\} \qquad p(M_1) = 0.5$$
$$\pi_1(\theta) = \begin{cases} 1 & si \quad \theta = 0.5 \\ 0 & otherwise \end{cases}$$

$$H_2 : M_2 = \{p(\tilde{y} \mid \theta) = Binomial(n, \theta), \theta \in (0,1)\} \qquad p(M_2) = 0.5$$
$$\pi_2(\theta) = Beta(1,1)$$

2.5 Hypothesis Test

Example 4: Simple vs Composite (continued)

We toss the coin $n = 200$ times and observe $y = 115$ heads.

We want to calculate the posterior probability for each hypothesis:

$$p(M_i \mid y) = \frac{p(M_i)p(y \mid M_i)}{\sum_{j=1}^{2} p(M_j)p(y \mid M_j)}$$

We get:

$$p(y \mid M_1) = \binom{200}{115} 0.5^{115}(1-0.5)^{200-115} = 0.00595$$

$$p(y \mid M_2) = \int_0^1 \binom{200}{115} \theta^{115}(1-\theta)^{200-115} d\theta = 0.00497$$

$$p(M_1 \mid y) = 0.54 \qquad FB_{12} = \frac{p(y \mid M_1)}{p(y \mid M_2)} = \frac{p(M_2)}{p(M_1)} \frac{p(M_1 \mid y)}{p(M_2 \mid y)} = 1.2$$
$$p(M_2 \mid y) = 0.46$$

We choose H_1

2.5 Hypothesis Test

Model Averaging Prediction

Given k models:

$$H_1 : M_1 = \{p_1(y \mid \theta_1), \theta_1 \in \Omega_1\}; \quad \pi_1(\theta_1); \quad p(M_1)$$
$$H_2 : M_2 = \{p_2(y \mid \theta_2), \theta_2 \in \Omega_2\}; \pi_2(\theta_2); \quad p(M_2)$$
$$\vdots$$
$$H_k : M_k = \{p_k(y \mid \theta_k), \theta_k \in \Omega_k\}; \pi_k(\theta_k); \quad p(M_k)$$

with posterior probabilities:

$$p(M_i \mid y) = \frac{p(M_i)p(y \mid M_i)}{\sum_{j=1}^{k} p(M_j)p(y \mid M_j)}$$

The model averaging consists of using a weighed average of the posterior predictive distributions of all models to make predictions:

$$p(\tilde{y} \mid y) = \sum_{j=1}^{k} p(M_j \mid y) p_j(\tilde{y} \mid y)$$

Chapter 2. Bayesian Inference

2.1 Posterior distribution as an estimator

2.2 Point Estimation

2.3 Interval Estimation

2.4 Prediction

2.5 Hypothesis Test

Chapter 2
Bayesian Inference

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